CBET / Biomed Academy

Technical References & Review Status

Prepared by MedSkillBuilder LLC | October 14, 2026

Resource: 10-mission biomedical equipment learning academy.

Source-map preparation date: October 14, 2026. Mission topics were matched to the selected references below. This date is not a claim of a completed question-by-question technical validation.

Independent review status: BMET/HTM volunteer review in progress. Feedback from the two selected reviewers has not yet been returned; no independent approval or endorsement is claimed.

Purpose

This packet helps learners and evaluating organizations locate primary technical and educational resources relevant to the Academy. The mission map identifies topic support and practical limits. It is not a service manual, certification, accreditation, compliance determination or proof of workplace competency.

How to use the references

Textbook and protocol references support general learning. Manufacturer documents are examples for the identified product or family. For real equipment, use its current model-specific instructions, the applicable standard edition, approved facility procedures and the authorization appropriate to your role.

Cited organizations are not represented as sponsors, approvers or endorsers. References are attributed by publisher, title and link; external manuals, illustrations and standards are not reproduced.

Mission-by-mission source map

Mission 1: Electronics Fundamentals

Topic scope: Voltage, current, resistance, Ohm's law, power and circuit behavior.

Mapped references: R1, R2

Application limit: Use the physics references for theory; work on real equipment requires appropriate training and approved safe procedures.

Mission 2: Electronic Components

Topic scope: Resistors, capacitors, diodes and component behavior.

Mapped references: R1, R2, R3, R4

Application limit: Transistor, relay, transformer and component-failure examples additionally require the applicable datasheet or schematic. The listed sources do not validate every component scenario.

Mission 3: Test Equipment

Topic scope: Meter setup, oscilloscopes, safety analyzers and simulator concepts.

Mapped references: R5, R6, R7

Application limit: Patient simulators, infusion/defibrillator analyzers and pressure/flow/temperature instruments require their own current manuals, calibration records and approved procedures.

Mission 4: Medical Equipment

Topic scope: Monitoring, infusion, defibrillation, ventilation, surgery, anesthesia and imaging systems.

Mapped references: R7, R8, R9, R16, R17

Application limit: These are selected device-category and maintenance references. Exact operation, accessories, PM and troubleshooting requirements come from the manufacturer for the actual model.

Mission 5: Anatomy & Physiology

Topic scope: Body systems and physiological signals used by clinical equipment.

Mapped references: R10

Application limit: Anatomy learning is not authorization to assess or treat a patient.

Mission 6: Networking & Computers

Topic scope: DHCP, DNS, medical-image exchange, clinical messages and cybersecurity.

Mapped references: R11, R12, R13, R14, R15

Application limit: TCP/IP, VLAN, Wi-Fi, PACS and interface configurations require approved IT documentation. Do not change clinical networks, patch devices or use patient data without authorization.

Mission 7: Safety & Regulations

Topic scope: Electrical safety, equipment management, recalls, documentation and reporting.

Mapped references: R7, R16, R17, R18, R19, R20, R21

Application limit: Use the applicable standard edition, manufacturer instructions and facility policy. A manufacturer analyzer manual does not replace a full standard or establish one universal leakage limit.

Mission 8: Troubleshooting

Topic scope: Separate device, accessory, setup, power and network evidence; verify repairs.

Mapped references: R5, R7, R15, R16, R17, R18, R20

Application limit: The sources support safety and maintenance context. Academy fault scenarios are instructional examples, not validated service procedures for named clinical models.

Mission 9: Field Troubleshooting Decisions

Topic scope: Complaint analysis, signal paths, interlocks and return-to-service decisions.

Mapped references: R7, R15, R16, R17, R18, R20, R21

Application limit: A correct quiz response or successful simulated repair does not establish competence to service equipment independently.

Mission 10: Medical Gas Delivery Equipment

Topic scope: Cylinder connections, regulators, flowmeters, blenders and suction setup.

Mapped references: R22, R23, R24, R25

Application limit: Connection types, gas identifiers, supply pressures and tests are model- and location-specific. Use current gas-specific IFUs and facility procedures; pipeline/zone work belongs to authorized qualified personnel.

Master bibliography

Primary-source links selected October 14, 2026. For web resources without a fixed edition, consult the publisher's current page. Dated sources and example manuals are identified below; publication age alone does not establish applicability to a particular device.

R1. OpenStax. University Physics Volume 2: 9.4 Ohm's Law

Voltage, current, resistance; foundational circuit relationships.

Open source on publisher website

R2. OpenStax. University Physics Volume 2: 10.2 Resistors in Series and Parallel

Series/parallel behavior and resistance calculations.

Open source on publisher website

R3. OpenStax. University Physics Volume 2: 8.1 Capacitors and Capacitance

Charge storage and capacitance fundamentals.

Open source on publisher website

R4. Texas Instruments. Diode Basics

Semiconductor diode behavior; introductory component learning.

Open source on publisher website

R5. Fluke. How to Measure Resistance with a Digital Multimeter

De-energized resistance measurement and meter setup.

Open source on publisher website

R6. Tektronix. XYZs of Oscilloscopes Primer

Waveforms, controls, time/voltage scales and measurement principles.

Open source on publisher website

R7. Fluke Biomedical. ESA615 Electrical Safety Analyzer Users Manual

Analyzer setup, defined electrical-safety test modes and ECG simulation. Model-specific example; limits are not universal.

Open source on publisher website

R8. FDA. Infusion Pumps

Device category, delivery functions and safety context; not a pump service procedure.

Open source on publisher website

R9. FDA. Ultrasound Imaging

Ultrasound imaging principles and safety context; not a model-specific service manual.

Open source on publisher website

R10. OpenStax. Anatomy and Physiology 2e

Body-system foundations; consult cardiovascular, respiratory, nervous and renal chapters for corresponding mission topics.

Open source on publisher website

R11. RFC Editor / IETF. RFC 2131: Dynamic Host Configuration Protocol

DHCP addressing and configuration. Foundational 1997 RFC; the record links to updates.

Open source on publisher website

R12. RFC Editor / IETF. RFC 1034: Domain Names - Concepts and Facilities

DNS concepts and name resolution. Foundational 1987 RFC; consult listed updates when implementing systems.

Open source on publisher website

R13. DICOM / NEMA. DICOM PS3.1: Introduction and Overview

Medical imaging information exchange and interoperability context.

Open source on publisher website

R14. HL7 International. HL7 Version 2 Product Suite

Clinical message exchange overview; not a hospital interface configuration specification.

Open source on publisher website

R15. FDA. Cybersecurity

Medical-device cybersecurity resources and safety communications.

Open source on publisher website

R16. CMS. Hospital Equipment Maintenance Requirements, S&C 14-07-Hospital (2013)

Risk-based equipment maintenance and documentation, including restrictions on alternate equipment management.

Open source on publisher website

R17. FDA. Remanufacturing of Medical Devices: Final Guidance (May 2024)

Distinction between servicing and changes affecting device specifications or intended use.

Open source on publisher website

R18. OSHA. 29 CFR 1910.147: Control of Hazardous Energy

Hazardous-energy control. Applicability and exceptions matter; this is not a universal procedure for every device.

Open source on publisher website

R19. OSHA. 29 CFR 1910.303: General Electrical Requirements

Workplace electrical installation and equipment safety context.

Open source on publisher website

R20. FDA. Medical Device Recalls and Early Alerts

Recall identification, affected equipment and manufacturer corrective actions.

Open source on publisher website

R21. FDA. Mandatory Reporting Requirements: Manufacturers, Importers and Device User Facilities

Reporting responsibilities and escalation context; institutional roles differ.

Open source on publisher website

R22. OSHA. 29 CFR 1910.101: Compressed Gases

Cylinder inspection and compressed-gas handling requirements.

Open source on publisher website

R23. Precision Medical. Air-Oxygen Blender Operating Instructions, document 504745, Rev. 24

Two gas supplies, alarm checks, oxygen-concentration verification and flowmeter orientation; model-specific.

Open source on publisher website

R24. Precision Medical. Dial Regulator User Manual, document 502444, Rev. 11

Cylinder connection, seals, flow settings and oxygen contamination hazards; model-specific.

Open source on publisher website

R25. Ohio Medical. Thoracic Vacuum Regulator Instructions for Use, document 6700-0358-000, Rev. 12 (June 2024)

Suction/vacuum setup and accessories; thoracic model example, not instructions for all suction regulators.

Open source on publisher website

Review status & appropriate use

What this packet establishes

The Academy's 10 mission titles and selected lesson topics have been organized into a source map. This is a bibliography and topic-mapping exercise, not a full accuracy audit or independent professional validation. Broad references do not automatically substantiate every quiz answer or fault scenario.

What remains in progress

Independent BMET/HTM volunteer feedback remains pending. Reviewer names, credentials, review scope and findings should be published only after the work is completed and disclosure is authorized.

Practical boundaries

Simulated service calls, XP, passing scores and certificates represent educational progress. They do not grant authorization to maintain patient-care equipment, alter clinical networks, perform medical-gas pipeline work or override manufacturer restrictions.

Do not infer universal electrical-safety limits, gas pressures, connector colors, test configurations or maintenance intervals from a general lesson or an example manual. Institution-specific requirements and applicable law or standards must be assessed by qualified personnel.

Report a correction

Identify the mission, lesson or question, explain the concern, and include a relevant supporting source when available.

MedSkillBuilder contact page

This packet publishes no fixed update schedule.